Element C

C1 (Determination of design requirements):

- Withstand a weight range of (inclinometer weight average)
- Should be no smaller than 2.36"L x 2.2" W
- Should have laser covers on both sides of the device
- The stand should be adjustable to fit in a storage room
- Should be smaller than 18 inches to be able to fit in the cupboard
- The feet of the stand must have the most effective shape for stability
- Must be able to sit on linoleum ground
- Must be easily movable for daily usage
- Must be able to work for>1 year
- Materials must be reasonably priced with a total price of ≈ $60
- Must be able to set it up and take it down due to limited class time in 3-5 minutes

C2 and C3 (Requirements list and priority ranking):

Priority	**Design requirement**	**Explanation**	**Source**	**What is measured (C4)**
High Priority (1)	The device must be able to do its job (connect 2 points on a wall and measure the height)	Meaning it should do its job	Primary Stakeholder (Dr.K)	Angle of measurement, accuracy of the final product
High priority (2)	Laser safety is connected to the device.	The lasers should have a shield in front of them or a way to protect the eye areas. We are considering adding some hinged doors that are connected to the box idea.	Primary Stakeholder (Forensic Students)	Are the lasers obstructed when in use? Is the safety device relatively easy to activate or deactivate? The laser doors should be a maximum of 3mm x 2 mm
High priority (3)	The prototype needs to be stable.	Meaning that it shouldn't fluctuate between angle measurements. We are considering creating a close-fitting box for the device with magnets attached at the top and bottom to keep the device within the box.	Primary Stakeholder (Dr. K)	How much does the device move when handled? What is the range of the angle reading? The box in which the inclinometer is held is going to be approximately the dimensions of the device: 2.36" x 2.2"
High priority (4)	Must be adjustable (up and down) and able to rotate 360 degrees.	The priority should be adjustability. Creating a rotating device (rod) at the back of the box	Primary stakeholder (Dr. K)	Can the device rotate 360 degrees? We could create Can the device connect two points to

		construction. Additionally, the box will be mounted to an adjustable system to ensure it can be raised and lowered.		produce a readable angle?
High priority (5)	The device is damage-proof	This device needs to be damage-proof, so when bumped, it must not break on impact.	Primary Stakeholder (Forensic Students)	Does it have any significant scrapes or dents after being dropped 5 times?
High priority (6)	Should be able to deliver prototype by May 19	This means that we as a group should be able to deliver a prototype by a certain deadline, which in this case is	Primary stakeholder (Dr.K and other forensic teachers)	Can it be delivered by this time?
Medium/low priority (6)	Should say "Manufactured by RMHS."	Meaning anywhere around the device, it must say that RMHS has manufactured it	Primary Stakeholder (Dr. K)	size/location of letters
Medium/low priority (7)	Able to fit in a cupboard/drawer	This device should be compact enough to be able to fit in a drawer or cupboard.	Primary Stakeholder (Dr. K) / Forensic Students	Should fit in a standard classroom drawer/cupboard. [ask Dr. K for dimensions]
Lower priority (8)	Can be Colorful (RMHS colors like Yellow and Black), mainly darker colors.	Meaning that the device/gadget itself should be like some sort of dark shades of colors, bright colors seem cheap. For example, bright colors like orange, yellow, and pink seem cheap	Primary Stakeholder (Dr. K)	Intensity of color, saturation of color. Specific hue?

C6:

Dr K, our primary stakeholder, cleared all of these priorities and offered some constructive advice.

We sent a survey out to Dr. K to relay to his students, but we have not yet received any replies.